

6.2 INFORMATION FILTERING

In this section, we discuss information filtering. This is an implementation of the Kalman filter that propagates the inverse of P rather than propagating P ; that is, information filtering propagates the information matrix of the system. Recall that

$$P = E[(x - \hat{x})(x - \hat{x})^T] \quad (6.23)$$

That is, P represents the uncertainty in the state estimate. If P is “large” then we have a lot of uncertainty in our state estimate. In the limit as $P \rightarrow 0$ we have perfect knowledge of x , and as $P \rightarrow \infty$ we have zero knowledge of x . The information matrix is defined as

$$\mathcal{I} = P^{-1} \quad (6.24)$$

That is, $\mathcal{I}$ represents the certainty in the state estimate. If $\mathcal{I}$ is “large” then we have a lot of confidence in our state estimate. In the limit as $\mathcal{I} \rightarrow 0$ we have zero knowledge of x , and as $\mathcal{I} \rightarrow \infty$ we have perfect knowledge of x .

Recall from Equation (5.19) that the measurement update equation for P can be written as

$$(P_k^+)^{-1} = (P_k^-)^{-1} + H_k^T R_k^{-1} H_k \quad (6.25)$$

Substituting the definition of $\mathcal{I}$ into this equation gives

$$\mathcal{I}_k^+ = \mathcal{I}_k^- + H_k^T R_k^{-1} H_k \quad (6.26)$$

This gives the measurement-update equation for the information matrix. Recall from Equation (5.19) the time-update equation for P :

$$P_k^- = F_{k-1} P_{k-1}^+ F_{k-1}^T + Q_{k-1} \quad (6.27)$$

This implies that

$$\mathcal{I}_k^- = [F_{k-1}(\mathcal{I}_{k-1}^+)^{-1} F_{k-1}^T + Q_{k-1}]^{-1} \quad (6.28)$$

Now we can use the matrix inversion lemma from Section 1.1.2, which we restate here:

$$(A + BD^{-1}C)^{-1} = A^{-1} - A^{-1}B(D + CA^{-1}B)^{-1}CA^{-1} \quad (6.29)$$

If we make the identifications $A = Q_{k-1}$, $B = F_{k-1}$, $C = F_{k-1}^T$, and $D = \mathcal{I}_{k-1}^+$, then we can apply the matrix inversion lemma to Equation (6.28) to obtain

$$\mathcal{I}_k^- = Q_{k-1}^{-1} - Q_{k-1}^{-1} F_{k-1} (\mathcal{I}_{k-1}^+ + F_{k-1}^T Q_{k-1}^{-1} F_{k-1})^{-1} F_{k-1}^T Q_{k-1}^{-1} \quad (6.30)$$

This gives the time-update equation for the information matrix. The information filter can be summarized as follows.

The information filter

1. The dynamic system is given by the following equations:

$$\begin{aligned} x_k &= F_{k-1} x_{k-1} + G_{k-1} u_{k-1} + w_{k-1} \\ y_k &= H_k x_k + v_k \\ w_k &\sim (0, Q_k) \end{aligned}$$

$$\begin{aligned}
v_k &\sim (0, R_k) \\
E(w_k w_j^T) &= Q_k \delta_{k-j} \\
E(v_k v_j^T) &= R_k \delta_{k-j} \\
E(w_k v_k^T) &= 0
\end{aligned} \tag{6.31}$$

2. The Kalman filter is initialized as follows:

$$\begin{aligned}
\hat{x}_0^+ &= E(x_0) \\
\mathcal{I}_0^+ &= \{E[(x_0 - \hat{x}_0^+)(x_0 - \hat{x}_0^+)^T]\}^{-1}
\end{aligned} \tag{6.32}$$

3. The information filter is given by the following equations, which are computed for each time step $k = 1, 2, \dots$:

$$\begin{aligned}
\mathcal{I}_k^- &= Q_{k-1}^{-1} - Q_{k-1}^{-1} F_{k-1} (\mathcal{I}_{k-1}^+ + F_{k-1}^T Q_{k-1}^{-1} F_{k-1})^{-1} F_{k-1}^T Q_{k-1}^{-1} \\
\mathcal{I}_k^+ &= \mathcal{I}_k^- + H_k^T R_k^{-1} H_k \\
K_k &= (\mathcal{I}_k^+)^{-1} H_k^T R_k^{-1} \\
\hat{x}_k^- &= F_{k-1} \hat{x}_{k-1}^+ + G_{k-1} u_{k-1} \\
\hat{x}_k^+ &= \hat{x}_k^- + K_k (y_k - H_k \hat{x}_k^-)
\end{aligned} \tag{6.33}$$

The standard Kalman filter equations require the inversion of an $r \times r$ matrix, where r is the number of measurements. The information filter equations require at least a couple of $n \times n$ matrix inversions, where n is the number of states. Therefore, if $r \gg n$ (i.e., we have significantly more measurements than states) it may be computationally more efficient to use the information filter. It could be argued that since the Kalman gain is given as

$$K_k = P_k^+ H_k^T R_k^{-1} \tag{6.34}$$

we have to perform and $r \times r$ matrix inversion on R_k anyway, whether we use the standard Kalman filter or the information filter. But if R_k is constant, then we could invert it as part of the initialization process, so the Kalman gain equation may not require this $r \times r$ matrix inversion after all. The same thinking also applies to the inversion of Q_{k-1} .

If the initial uncertainty is infinite, we cannot numerically set $P_0^+ = \infty$, but we can numerically set $\mathcal{I}_0^+ = 0$. This makes the information filter more mathematically precise for the zero initial certainty case. However, if the initial uncertainty is zero (i.e., we have perfect knowledge of x_0), we can numerically set $P_0^+ = 0$, but we cannot numerically set $\mathcal{I}_0^+ = \infty$. This makes the standard Kalman filter more mathematically precise for the zero initial uncertainty case

■ EXAMPLE 6.2

The information filter can be used to solve the American football team ranking problem of Example 6.1. The information filter equations are given as

$$\begin{aligned}
\mathcal{I}_1^- &= Q^{-1} - Q^{-1} F (\mathcal{I}_0^+ + F^T Q^{-1} F)^{-1} F^T Q^{-1} \\
&= 0.1783
\end{aligned}$$